

SOLIDWORKS API (.NET VBA C#) — Reference (159/337): IMeasure-IMessageBarDefinition

SOLIDWORKS API Documentation (converted from CHM)

IMeasure (Type)

IMeasure Interface

[See Also](#) [Members](#) [Example](#)

 

[SolidWorks.Interop.sldworks Namespace](#) : IMeasure Interface

Allows access to the measure tool.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

 **.NET Syntax**

Visual Basic (Declaration)

Public Interface IMeasure

Visual Basic (Usage)

Dim instance As [IMeasure](#)

C#

public interface IMeasure

C++/CLI

public interface class IMeasure

 **Visual Basic for Applications (VBA) Syntax**

See [Measure](#).

 **Example**

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

 **Accessors**

[IModelDocExtension::CreateMeasure](#)

 **Access Diagram**

[Measure](#)

 **See Also**

[IMeasure Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

IMeasure Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMeasure Interface

The following tables list the members exposed by [IMeasure](#).

 **Public Properties**

Name	Description
 Angle	Gets the angle between the two selected entities.
 AngleDecimalPlaces	Gets or sets the number of decimal places for angle measurements.

 ArcLength	Gets the length of the selected arc.
 ArcOption	Gets or sets how to measure an arc or circle.
 Area	Gets the area of the selected entity.
 CenterDistance	Gets the distance between the selected cylinder axes, circle centers, or both.
 ChordLength	Gets the chord length of the selected arc.
 DeltaX	Gets the Delta X distance between the selected entities.
 DeltaY	Gets the Delta Y distance between the selected entities.
 DeltaZ	Gets the Delta Z distance between the selected entities.
 Diameter	Gets the diameter of the selected circle or cylinder.
 Distance	Gets the distance between the selected entities.
 IsConcentricSpheres	Gets whether the two selected entities are concentric spheres.
 IsIntersect	Gets whether the two selected entities intersect.
 IsParallel	Gets whether the two selected entities are parallel.
 IsPerpendicular	Gets whether the two selected entities are perpendicular.
 Length	Gets the length of the selected entity.
 LengthDecimalPlaces	Gets or sets the number of decimal places for length measurements.
 Normal	Gets the normal distance (normal to the selected plane) between the selected face and plane.
 NormalDistance	Gets the normal distance (normal to the selected plane) between the selected entity and plane.
 Perimeter	Gets the perimeter of the selected entity.
 Projection	Gets the projected distance between the two selected entities.
 ProjectionOption	Gets or sets whether the distance between the selected entities is projected.
 Radius	Gets the radius of the selected arc or cylinder.
 SphericalCenterDistance	Gets the distance between the two selected spherical faces.
 TotalArea	Gets the total area of the two selected entities.
 TotalLength	Gets the total length of the two selected entities.
 X	Gets the x coordinate of the selected point in model space.
 Y	Gets the y coordinate of the selected point in model space.
 Z	Gets the z coordinate of the selected point in model space.

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Public Methods

Name	Description
 Calculate	Measures the selected or specified entities.
 SetProjectionEntity	Sets the face or plane to which to project the measured distance.

[Top](#)

See Also

[IMeasure Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

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IMeasure Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMeasure Interface

For a list of all members of this type, see [IMeasure members](#).

Public Methods

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 Calculate	Measures the selected or specified entities.
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[Top](#)

See Also

[IMeasure Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

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IMeasure Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMeasure Interface

For a list of all members of this type, see [IMeasure members](#).

Public Properties

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See Also

[IMeasure Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

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Angle Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : Angle Property (IMeasure)

Gets the angle between the two selected entities.

Visual Basic Syntax

Visual Basic (Declaration)

ReadOnly Property Angle As System.Double

Visual Basic (Usage)



```
Dim instance As IMeasure
Dim value As System.Double

value = instance.Angle
```

C#

```
System.double Angle {get;}
```

C++/CLI

```
property System.double Angle {
    System.double get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Angle in radians between the two selected entities or -1 if this property is invalid for the selected entities

Visual Basic for Applications (VBA) Syntax

See [Measure::Angle](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

See Also

[IMeasure Interface](#)
[IMeasure Members](#)
[IMeasure::AngleDecimalPlaces Property](#)

Availability

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AngleDecimalPlaces Property (IMeasure)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : AngleDecimalPlaces Property (IMeasure)

Gets or sets the number of decimal places for angle measurements.

.NET Syntax

Visual Basic (Declaration)

```
Property AngleDecimalPlaces As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IMeasure
Dim value As System.Integer

instance.AngleDecimalPlaces = value

value = instance.AngleDecimalPlaces
```

C#

```
System.int AngleDecimalPlaces {get; set;}
```

C++/CLI

```
property System.int AngleDecimalPlaces {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Number of decimal places for angle measurements

Visual Basic for Applications (VBA) Syntax

See [Measure::AngleDecimalPlaces](#).

See Also

[IMeasure Interface](#)
[IMeasure Members](#)
[IMeasure::Angle Property](#)

Availability

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ArcLength Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : ArcLength Property (IMeasure)

Gets the length of the selected arc.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property ArcLength As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMeasure  
Dim value As System.Double  
  
value = instance.ArcLength
```

C#

```
System.double ArcLength {get;}
```

C++/CLI

```
property System.double ArcLength {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Length of arc or -1 if this property is invalid for the selected entity

Visual Basic for Applications (VBA) Syntax

See [Measure::ArcLength](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

See Also

[IMeasure Interface](#)
[IMeasure Members](#)
[IMeasure::ArcOption Property](#)

Availability

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ArcOption Property (IMeasure)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : ArcOption Property (IMeasure)

Gets or sets how to measure an arc or circle.

.NET Syntax

Visual Basic (Declaration)

```
Property ArcOption As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IMeasure  
Dim value As System.Integer  
  
instance.ArcOption = value  
  
value = instance.ArcOption
```

C#

```
System.int ArcOption {get; set;}
```

C++/CLI

```
property System.int ArcOption {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

How to measure an arc or circle as defined by [swMeasureArcCircleOption](#) e

Visual Basic for Applications (VBA) Syntax

See [Measure::ArcOption](#).

See Also

[IMeasure Interface](#)
[IMeasure Members](#)
[IMeasure::ArcLength Property](#)

Availability

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Area Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : Area Property (IMeasure)

Gets the area of the selected entity.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property Area As System.Double

Visual Basic (Usage)



```
Dim instance As IMeasure
Dim value As System.Double

value = instance.Area
```

C#

```
System.double Area {get;}
```

C++/CLI

```
property System.double Area {
    System.double get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Area or -1 if this property is invalid for the selected entity

Visual Basic for Applications (VBA) Syntax

See [Measure::Area](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

See Also

[IMeasure Interface](#)
[IMeasure Members](#)
[IMeasure::TotalArea Property](#)

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Calculate Method (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : Calculate Method (IMeasure)

Measures the selected or specified entities.

.NET Syntax

Visual Basic (Declaration)

```
Function Calculate( _  
    ByVal Entities As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMeasure  
Dim Entities As System.Object  
Dim value As System.Boolean  
  
value = instance.Calculate(Entities)
```

C#

```
System.bool Calculate(  
    System.object Entities  
)
```

C++/CLI

```
System.bool Calculate(  
& System.Object^ Entities  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Entities

Array of entities to measure; if NULL, then this method measures the selected entities

Return Value

True if the specified or selected entities are valid types and combination , false if not

Visual Basic for Applications (VBA) Syntax

See [Measure::Calculate](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

See Also

[IMeasure Interface](#)
[IMeasure Members](#)

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CenterDistance Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : CenterDistance Property (IMeasure)

Gets the distance between the selected cylinder axes, circle centers, or both.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property CenterDistance As System.Double

Visual Basic (Usage)



```
Dim instance As IMeasure
Dim value As System.Double

value = instance.CenterDistance
```

C#

```
System.double CenterDistance {get;}
```

C++/CLI

```
property System.double CenterDistance {
    System.double get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Distance between cylinder axes, circle centers, or both or -1 if this property is invalid for the selected entities

Visual Basic for Applications (VBA) Syntax

See [Measure::CenterDistance](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

See Also

[IMeasure Interface](#)
[IMeasure Members](#)

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ChordLength Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : ChordLength Property (IMeasure)

Gets the chord length of the selected arc.

.NET Syntax

```
ReadOnly Property ChordLength As System.Double
```



```
Dim instance As IMeasure
Dim value As System.Double

value = instance.ChordLength
```

```
System.double ChordLength {get;}
```

```
property System.double ChordLength {
    System.double get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Chord length or -1 if this property is invalid for the selected entity

Visual Basic for Applications (VBA) Syntax

See [Measure::ChordLength](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

See Also

[IMeasure Interface](#)
[IMeasure Members](#)

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DeltaX Property (IMeasure)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : DeltaX Property (IMeasure)

Gets the Delta X distance between the selected entities.

.NET Syntax

```
ReadOnly Property DeltaX As System.Double
```



```
Dim instance As IMeasure
Dim value As System.Double

value = instance.DeltaX
```

```
System.double DeltaX {get;}
```

C++/CLI

```
property System.double DeltaX {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Delta X distance between the selected entities or -1 if this property is invalid for the selected entities

Visual Basic for Applications (VBA) Syntax

See [Measure::DeltaX](#).

Example

See the [IMeasure](#) examples.

See Also

[IMeasure Interface](#)
[IMeasure Members](#)

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DeltaY Property (IMeasure)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : DeltaY Property (IMeasure)

Gets the Delta Y distance between the selected entities.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property DeltaY As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMeasure  
Dim value As System.Double
```

```
value = instance.DeltaY
```

C#

```
System.double DeltaY {get;}
```

C++/CLI

```
property System.double DeltaY {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Delta Y distance between the selected entities or -1 if this property is invalid for the selected entities

Visual Basic for Applications (VBA) Syntax

See [Measure::DeltaY](#).

Example

See the [IMeasure](#) examples.

See Also

[IMeasure Interface](#)
[IMeasure Members](#)

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DeltaZ Property (IMeasure)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : DeltaZ Property (IMeasure)

Gets the Delta Z distance between the selected entities.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property DeltaZ As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMeasure  
Dim value As System.Double
```

```
value = instance.DeltaZ
```

C#

```
System.double DeltaZ {get;}
```

C++/CLI

```
property System.double DeltaZ {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Delta Z distance between the selected entities or -1 if this property is invalid for the selected entities

Visual Basic for Applications (VBA) Syntax

See [Measure::DeltaZ](#).

Example

See the [IMeasure](#) examples.

See Also

[IMeasure Interface](#)
[IMeasure Members](#)

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Diameter Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : Diameter Property (IMeasure)

Gets the diameter of the selected circle or cylinder.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Diameter As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMeasure  
Dim value As System.Double  
  
value = instance.Diameter
```

C#

```
System.double Diameter {get;}
```

C++/CLI

```
property System.double Diameter {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Diameter of the selected circle or cylinder or -1 if this property is invalid for the selected entity

Visual Basic for Applications (VBA) Syntax

See [Measure::Diameter](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

See Also

[IMeasure Interface](#)
[IMeasure Members](#)
[IMeasure::Radius Property](#)

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Distance Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : Distance Property (IMeasure)

Gets the distance between the selected entities.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Distance As System.Double
```

Visual Basic (Usage)

```
Dim instance As IMeasure  
Dim value As System.Double  
  
value = instance.Distance
```

C#

```
System.double Distance {get;}
```

C++/CLI

```
property System.double Distance {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Distance between the selected entities or -1 if this property is invalid for the selected entities

Visual Basic for Applications (VBA) Syntax

See [Measure.Distance](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
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See Also

[IMeasure Interface](#)
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IsConcentricSpheres Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : IsConcentricSpheres Property (IMeasure)

Gets whether the two selected entities are concentric spheres.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property IsConcentricSpheres As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IMeasure  
Dim value As System.Boolean  
  
value = instance.IsConcentricSpheres
```

C#

```
System.bool IsConcentricSpheres {get;}
```

C++/CLI

```
property System.bool IsConcentricSpheres {  
    System.bool get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if the two selected entities are concentric spheres, false if not

Visual Basic for Applications (VBA) Syntax

See [Measure::IsConcentricSpheres](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

See Also

[IMeasure Interface](#)
[IMeasure Members](#)
[IMeasure::IsIntersect Property](#)
[IMeasure::IsParallel Property](#)
[IMeasure::IsPerpendicular Property](#)

Availability

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IsIntersect Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : IsIntersect Property (IMeasure)

Gets whether the two selected entities intersect.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property IsIntersect As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMeasure  
Dim value As System.Boolean  
  
value = instance.IsIntersect
```

C#

```
System.bool IsIntersect {get;}
```

C++/CLI

```
property System.bool IsIntersect {  
    System.bool get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if the two selected entities intersect, false if not

Visual Basic for Applications (VBA) Syntax

See [Measure::IsIntersect](#).

Example

[Measure Selected Entities \(C#\)](#)
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[IMeasure Interface](#)
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[IMeasure::IsConcentricSpheres Property](#)
[IMeasure::IsParallel Property](#)
[IMeasure::IsPerpendicular Property](#)

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IsParallel Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : IsParallel Property (IMeasure)

Gets whether the two selected entities are parallel.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property IsParallel As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMeasure  
Dim value As System.Boolean  
  
value = instance.IsParallel
```

C#

```
System.bool IsParallel {get;}
```

C++/CLI

```
property System.bool IsParallel {  
    System.bool get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if the two selected entities are parallel, false if not

Visual Basic for Applications (VBA) Syntax

See [Measure::IsParallel](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

See Also

[IMeasure Interface](#)
[IMeasure Members](#)
[IMeasure::IsConcentricSpheres Property](#)

[IMeasure::IsIntersect Property](#)
[IMeasure::IsPerpendicular Property](#)

Availability

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IsPerpendicular Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : IsPerpendicular Property (IMeasure)

Gets whether the two selected entities are perpendicular.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property IsPerpendicular As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMeasure  
Dim value As System.Boolean  
  
value = instance.IsPerpendicular
```

C#

```
System.bool IsPerpendicular {get;}
```

C++/CLI

```
property System.bool IsPerpendicular {  
    System.bool get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if the two selected entities are perpendicular, false if not

Visual Basic for Applications (VBA) Syntax

See [Measure::IsPerpendicular](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

See Also

[IMeasure Interface](#)
[IMeasure Members](#)
[IMeasure::IsConcentricSpheres Property](#)
[IMeasure::IsIntersect Property](#)
[IMeasure::IsParallel Property](#)

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Length Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : Length Property (IMeasure)

Gets the length of the selected entity.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Length As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMeasure  
Dim value As System.Double  
  
value = instance.Length
```

C#

```
System.double Length {get;}
```

C++/CLI

```
property System.double Length {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Length of the selected entity or -1 if this property is invalid for the selected entity

Visual Basic for Applications (VBA) Syntax

See [Measure::Length](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

See Also

[IMeasure Interface](#)
[IMeasure Members](#)
[IMeasure::LengthDecimalPlaces Property](#)
[IMeasure::TotalLength Property](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

LengthDecimalPlaces Property (IMeasure)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : LengthDecimalPlaces Property (IMeasure)

Gets or sets the number of decimal places for length measurements.

.NET Syntax

Visual Basic (Declaration)

```
Property LengthDecimalPlaces As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IMeasure  
Dim value As System.Integer
```

```
instance.LengthDecimalPlaces = value  
value = instance.LengthDecimalPlaces
```

C#

```
System.int LengthDecimalPlaces {get; set;}
```

C++/CLI

```
property System.int LengthDecimalPlaces {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Number of decimal places for length measurements

Visual Basic for Applications (VBA) Syntax

See [Measure::LengthDecimalPlaces](#).

See Also

[IMeasure Interface](#)
[IMeasure Members](#)
[IMeasure::Length Property](#)
[IMeasure::TotalLength Property](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Normal Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : Normal Property (IMeasure)

Gets the normal distance (normal to the selected plane) between the selected face and plane.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Normal As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMeasure  
Dim value As System.Double  
  
value = instance.Normal
```

C#

```
System.double Normal {get;}
```

C++/CLI

```
property System.double Normal {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Normal distance between the selected face and plane or -1 if this property is invalid for the selected entities

Visual Basic for Applications (VBA) Syntax

See [Measure::Normal](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

See Also

[IMeasure Interface](#)
[IMeasure Members](#)
[IMeasure::NormalDistance Property](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

NormalDistance Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : NormalDistance Property (IMeasure)

Gets the normal distance (normal to the selected plane) between the selected entity and plane.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property NormalDistance As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMeasure  
Dim value As System.Double  
  
value = instance.NormalDistance
```

C#

```
System.double NormalDistance {get;}
```

C++/CLI

```
property System.double NormalDistance {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Normal distance or -1 if this property is invalid for the selected entities

Visual Basic for Applications (VBA) Syntax

See [Measure::NormalDistance](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

See Also

[IMeasure Interface](#)
[IMeasure Members](#)
[IMeasure::Normal Property](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Perimeter Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : Perimeter Property (IMeasure)

Gets the perimeter of the selected entity.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Perimeter As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMeasure  
Dim value As System.Double  
  
value = instance.Perimeter
```

C#

```
System.double Perimeter {get;}
```

C++/CLI

```
property System.double Perimeter {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Perimeter of the selected entity or -1 if this property is invalid for the selected entity

Visual Basic for Applications (VBA) Syntax

See [Measure::Perimeter](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

See Also

[IMeasure Interface](#)
[IMeasure Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Projection Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : Projection Property (IMeasure)

Gets the projected distance between the two selected entities.

Visual Basic Syntax

Visual Basic (Declaration)

```
ReadOnly Property Projection As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMeasure
Dim value As System.Double

value = instance.Projection
```

C#

```
System.double Projection {get;}
```

C++/CLI

```
property System.double Projection {
    System.double get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Projected distance between the two selected entities or -1 if this property is invalid for the selected entities

Visual Basic for Applications (VBA) Syntax

See [Measure::Projection](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

See Also

[IMeasure Interface](#)
[IMeasure Members](#)
[IMeasure::ProjectionOption Property](#)
[IMeasure::SetProjectionEntity Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

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ProjectionOption Property (IMeasure)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : ProjectionOption Property (IMeasure)

Gets or sets whether the distance between the selected entities is projected.

Visual Basic Syntax

Visual Basic (Declaration)

```
Property ProjectionOption As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IMeasure
Dim value As System.Integer

instance.ProjectionOption = value
```

```
value = instance.ProjectionOption
```

C#

```
System.int ProjectionOption {get; set;}
```

C++/CLI

```
property System.int ProjectionOption {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Distance between the selected entities is projected as defined by [swMeasureProjectOnOption_e](#)

Visual Basic for Applications (VBA) Syntax

See [Measure::ProjectionOption](#).

See Also

[IMeasure Interface](#)
[IMeasure Members](#)
[IMeasure::SetProjectionEntity Method](#)
[IMeasure::Projection Property](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

Radius Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : Radius Property (IMeasure)

Gets the radius of the selected arc or cylinder.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Radius As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMeasure  
Dim value As System.Double  
  
value = instance.Radius
```

C#

```
System.double Radius {get;}
```

C++/CLI

```
property System.double Radius {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Radius

Visual Basic for Applications (VBA) Syntax

See [Measure::Radius](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

See Also

[IMeasure Interface](#)
[IMeasure Members](#)
[IMeasure::Diameter Property](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

SetProjectionEntity Method (IMeasure)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : SetProjectionEntity Method (IMeasure)

Sets the face or plane to which to project the measured distance.

.NET Syntax

Visual Basic (Declaration)

```
Function SetProjectionEntity( _  
    ByVal Entity As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMeasure  
Dim Entity As System.Object  
Dim value As System.Boolean  
  
value = instance.SetProjectionEntity(Entity)
```

C#

```
System.bool SetProjectionEntity(  
    System.object Entity  
)
```

C++/CLI

```
System.bool SetProjectionEntity(  
& System.Object^ Entity  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Entity
[Face](#) or [plane](#)

Return Value

True if Entity is set, false if not

Visual Basic for Applications (VBA) Syntax

See [Measure::SetProjectionEntity](#).

See Also

[IMeasure Interface](#)
[IMeasure Members](#)
[IMeasure::Projection Property](#)
[IMeasure::ProjectionOption Property](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SphericalCenterDistance Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : SphericalCenterDistance Property (IMeasure)

Gets the distance between the two selected spherical faces.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property SphericalCenterDistance As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMeasure  
Dim value As System.Double  
  
value = instance.SphericalCenterDistance
```

C#

```
System.double SphericalCenterDistance {get;}
```

C++/CLI

```
property System.double SphericalCenterDistance {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Distance between the two selected spherical faces or -1 if this property is invalid for the selected entities

Visual Basic for Applications (VBA) Syntax

See [Measure::SphericalCenterDistance](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

See Also

[IMeasure Interface](#)
[IMeasure Members](#)

Availability

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SOLIDWORKS API Help

[Send comments on this topic.](#)

TotalArea Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : TotalArea Property (IMeasure)

Gets the total area of the two selected entities.

Visual Basic Syntax

Visual Basic (Declaration)

```
ReadOnly Property TotalArea As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMeasure
Dim value As System.Double

value = instance.TotalArea
```

C#

```
System.double TotalArea {get;}
```

C++/CLI

```
property System.double TotalArea {
    System.double get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Total area of the two selected entities or -1 if this property is invalid for the selected entities

Visual Basic for Applications (VBA) Syntax

See [Measure::TotalArea](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

See Also

[IMeasure Interface](#)
[IMeasure Members](#)
[IMeasure::Area Property](#)

Availability

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SOLIDWORKS API Help

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TotalLength Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : TotalLength Property (IMeasure)

Gets the total length of the two selected entities.

Visual Basic Syntax

Visual Basic (Declaration)

```
ReadOnly Property TotalLength As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMeasure
Dim value As System.Double

value = instance.TotalLength
```

C#

```
System.double TotalLength {get;}
```

C++/CLI

```
property System.double TotalLength {
    System.double get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Total length of the two selected entities or -1 if this property is invalid for the selected entities

Visual Basic for Applications (VBA) Syntax

See [Measure::TotalLength](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

See Also

[IMeasure Interface](#)
[IMeasure Members](#)
[IMeasure::Length Property](#)
[IMeasure::LengthDecimalPlaces Property](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

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X Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : X Property (IMeasure)

Gets the x coordinate of the selected point in model space.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property X As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMeasure
Dim value As System.Double

value = instance.X
```

C#

```
System.double X {get;}
```

C++/CLI

```
property System.double X {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

x coordinate of the selected point in model space or -1 if this property is invalid for the selected entity

Visual Basic for Applications (VBA) Syntax

See [Measure::X](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

See Also

[IMeasure Interface](#)
[IMeasure Members](#)
[IMeasure::Y Property](#)
[IMeasure::Z Property](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Y Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : Y Property (IMeasure)

Gets the y coordinate of the selected point in model space.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Y As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMeasure  
Dim value As System.Double  
  
value = instance.Y
```

C#

```
System.double Y {get;}
```

C++/CLI

```
property System.double Y {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

y coordinate of the selected point in model space or -1 if this property is invalid for the selected entity

Visual Basic for Applications (VBA) Syntax

See [Measure::Y](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

See Also

[IMeasure Interface](#)
[IMeasure Members](#)
[IMeasure::X Property](#)
[IMeasure::Z Property](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Z Property (IMeasure)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeasure Interface](#) : Z Property (IMeasure)

Gets the z coordinate of the selected point in model space.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property Z As System.Double

Visual Basic (Usage)



```
Dim instance As IMeasure
Dim value As System.Double

value = instance.Z
```

C#

```
System.double Z {get;}
```

C++/CLI

```
property System.double Z {
    System.double get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

z coordinate of the selected point in model space or -1 if this property is invalid for the selected entity

Visual Basic for Applications (VBA) Syntax

See [Measure::Z](#).

Example

[Measure Selected Entities \(C#\)](#)
[Measure Selected Entities \(VB.NET\)](#)
[Measure Selected Entities \(VBA\)](#)

See Also

[IMeasure Interface](#)
[IMeasure Members](#)
[IMeasure::X Property](#)
[IMeasure::Y Property](#)

IMeshBody (Type)

IMeshBody Interface

[See Also](#) [Members](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMeshBody Interface

Allows access to a mesh body.
NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IMeshBody
```

Visual Basic (Usage)

```
Dim instance As IMeshBody
```

C#

```
public interface IMeshBody
```

C++/CLI

```
public interface class IMeshBody
```

Visual Basic for Applications (VBA) Syntax

See [MeshBody](#).

Example

See the [IFacet](#) examples.

Remarks

For more information about graphics bodies, read the **Graphics Mesh and Mesh BREP Bodies** topic in **SOLIDWORKS Design user-interface help > Parts and Features**.

Accessors

[IBody2::GetMeshBody](#)

Access Diagram

[MeshBody](#)

See Also

[IMeshBody Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IGraphicsBody Interface](#)

IMeshBody Interface Members

[See Also](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMeshBody Interface

The following tables list the members exposed by [IMeshBody](#).

Public Methods

Name	Description
------	-------------

 GetBody	Gets the body associated with this mesh body.
 GetFacetCount	Gets the number of facets on this mesh body.
 GetFacets	Gets the facets on this mesh body.
 SurfaceFromMesh	Creates a surface from the selected facets in this mesh body.

[Top](#)

 See Also

[IMeshBody Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IGraphicsBody Interface](#)

SOLIDWORKS API Help [Send comments on this topic.](#)

IMeshBody Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMeshBody Interface

For a list of all members of this type, see [IMeshBody members](#).

 Public Methods

Name	Description
 GetBody	Gets the body associated with this mesh body.
 GetFacetCount	Gets the number of facets on this mesh body.
 GetFacets	Gets the facets on this mesh body.
 SurfaceFromMesh	Creates a surface from the selected facets in this mesh body.

[Top](#)

 See Also

[IMeshBody Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IGraphicsBody Interface](#)

SOLIDWORKS API Help [Send comments on this topic.](#)

GetBody Method (IMeshBody)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeshBody Interface](#) : GetBody Method (IMeshBody)

Gets the body associated with this mesh body.

 .NET Syntax

Visual Basic (Declaration)

Function GetBody() As System.Object

Visual Basic (Usage)



```
Dim instance As IMeshBody  
Dim value As System.Object  
  
value = instance.GetBody()
```

C#

System.object GetBody()

C++/CLI

System.Object^ GetBody();

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IBody2](#)

Visual Basic for Applications (VBA) Syntax

See [MeshBody::GetBody](#).

Example

See the [IFacet](#) examples.

See Also

[IMeshBody Interface](#)
[IMeshBody Members](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetFacetCount Method (IMeshBody)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeshBody Interface](#) : GetFacetCount Method (IMeshBody)

Gets the number of facets on this mesh body.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFacetCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IMeshBody  
Dim value As System.Integer  
  
value = instance.GetFacetCount()
```

C#

```
System.int GetFacetCount()
```

C++/CLI

```
System.int GetFacetCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of facets

Visual Basic for Applications (VBA) Syntax

See [MeshBody::GetFacetCount](#).

Example

See the [IFacet](#) examples.

See Also

[IMeshBody Interface](#)
[IMeshBody Members](#)

[IMeshBody::GetFacets Method \(\)](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFacets Method (IMeshBody)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeshBody Interface](#) : GetFacets Method (IMeshBody)

Gets the facets on this mesh body.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFacets() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMeshBody  
Dim value As System.Object  
  
value = instance.GetFacets()
```

C#

```
System.object GetFacets()
```

C++/CLI

```
System.Object^ GetFacets();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [IFacets](#)

Visual Basic for Applications (VBA) Syntax

See [MeshBody::GetFacets](#).

Example

See the [IFacet](#) examples.

See Also

[IMeshBody Interface](#)
[IMeshBody Members](#)
[IMeshBody::GetFacetCount Method \(\)](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

SurfaceFromMesh Method (IMeshBody)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMeshBody Interface](#) : SurfaceFromMesh Method (IMeshBody)

Creates a surface from the selected facets in this mesh body.

.NET Syntax

```

Function SurfaceFromMesh( _
    ByVal SurfaceType As System.Integer, _
    ByVal FacetTolerance As System.Double, _
    ByVal ExtendSurfaceSize As System.Double, _
    ByRef ErrorCode As System.Integer _
) As System.Object

```

```

Dim instance As IMeshBody
Dim SurfaceType As System.Integer
Dim FacetTolerance As System.Double
Dim ExtendSurfaceSize As System.Double
Dim ErrorCode As System.Integer
Dim value As System.Object

value = instance.SurfaceFromMesh(SurfaceType, FacetTolerance, ExtendSurfaceSize, ErrorCode)

```

```

System.object SurfaceFromMesh(
    System.int SurfaceType,
    System.double FacetTolerance,
    System.double ExtendSurfaceSize,
    out System.int ErrorCode
)

```

```

System.Object^ SurfaceFromMesh(
    & System.int SurfaceType,
    & System.double FacetTolerance,
    & System.double ExtendSurfaceSize,
    & [Out] System.int ErrorCode
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SurfaceType

Type of surface to create as defined by [swSurfaceTypes_e](#) (see **Remarks**)

FacetTolerance

0.0 (High tolerance) <= Percent facet selection <= 100.0 (Low tolerance)

ExtendSurfaceSize

Extension in meters of the surface body to fit the selected facets and geometric shape

ErrorCode

0 if successful, -1 if not

Return Value

[IBody2](#)

Visual Basic for Applications (VBA) Syntax

See [MeshBody::SurfaceFromMesh](#).

Example

See the [IBody2::ConvertToMeshBody](#) examples.

Remarks

This method:

- is analogous to the **Insert menu > Mesh > Surface From Mesh** functionality in the user interface.
- creates a **Surface-From-MeshX** feature in the FeatureManager design tree.

Before calling this method, select as many facets in the mesh body as are needed to create the type of surface.

SurfaceType can be one of only swSurfaceTypes_e.:

- PLANE_TYPE
- SPHERE_TYPE
- CYLINDER_TYPE
- CONE_TYPE

For more information, see the **SOLIDWORKS Design user-interface help > Parts and Features > Graphics Mesh and Mesh BREP Bodies : Creating a Surface from Mesh Feature** topic.

See Also

[IMeshBody Interface](#)
[IMeshBody Members](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

IMessageBarDefinition (Type)

IMessageBarDefinition Interface

[See Also](#) [Members](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMessageBarDefinition Interface

Allows access to a message bar in an add-in.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

Public Interface IMessageBarDefinition

Visual Basic (Usage)

Dim instance As [IMessageBarDefinition](#)

C#

public interface IMessageBarDefinition

C++/CLI

public interface class IMessageBarDefinition

Visual Basic for Applications (VBA) Syntax

See [MessageBarDefinition](#).

Example

See the [IMessageBarHandler](#) examples.

Remarks

A message bar:

- Allows an add-in to display a message to users.
- Is a modeless dialog that contains a unique message.
- Only appears within the context of a document open in its own frame.
- May be stacked with other message bars in the user interface.
- Requires an IMessageBarHandler to handle message bar events.

Accessors

[ISldWorks::DefineMessageBar](#)

Access Diagram

[MessageBarDefinition](#)

See Also

[IMessageBarDefinition Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IUserNotificationDefinition Interface](#)

IMessageBarDefinition Interface Members

[See Also](#) [Properties](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMessageBarDefinition Interface

The following tables list the members exposed by [IMessageBarDefinition](#).

Public Properties

Name	Description
 IncludeDoNotShowAgain	Gets or sets whether to display the "Do not show again" check box for this message bar.
 Message	Gets or sets the message displayed on this message bar.
 ResponseAText	Gets or sets the text displayed on the first button or command link of this message bar.
 ResponseAType	Gets or sets the first response user control for this message bar.
 ResponseBText	Gets or sets the text displayed on the second button or command link of this message bar.
 ResponseBType	Gets or sets the second response user control for this message bar.
 Severity	Gets or sets the severity level of this message bar.
 Title	Gets or sets the title of this message bar.
 UniqueName	Gets the ID of this message bar.

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See Also

[IMessageBarDefinition Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IUserNotificationDefinition Interface](#)

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IMessageBarDefinition Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMessageBarDefinition Interface

For a list of all members of this type, see [IMessageBarDefinition members](#).

Public Properties

Name	Description
 IncludeDoNotShowAgain	Gets or sets whether to display the "Do not show again" check box for this message bar.
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 Severity	Gets or sets the severity level of this message bar.
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 UniqueName	Gets the ID of this message bar.

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See Also

[IMessageBarDefinition Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IUserNotificationDefinition Interface](#)

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IncludeDoNotShowAgain Property (IMessageBarDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMessageBarDefinition Interface](#) : IncludeDoNotShowAgain Property (IMessageBarDefinition)

Gets or sets whether to display the "Do not show again" check box for this message bar.

.NET Syntax

Visual Basic (Declaration)

Property IncludeDoNotShowAgain As System.Boolean

Visual Basic (Usage)



```
Dim instance As IMessageBarDefinition
Dim value As System.Boolean

instance.IncludeDoNotShowAgain = value

value = instance.IncludeDoNotShowAgain
```

C#

```
System.bool IncludeDoNotShowAgain {get; set;}
```

C++/CLI

```
property System.bool IncludeDoNotShowAgain {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True (default) to display the check box, false to not.

Visual Basic for Applications (VBA) Syntax

See [MessageBarDefinition::IncludeDoNotShowAgain](#).

See Also

[IMessageBarDefinition Interface](#)
[IMessageBarDefinition Members](#)

Availability

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Message Property (IMessageBarDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMessageBarDefinition Interface](#) : Message Property (IMessageBarDefinition)

Gets or sets the message displayed on this message bar.

.NET Syntax

Visual Basic (Declaration)

Property Message As System.String

Visual Basic (Usage)



```
Dim instance As IMessageBarDefinition
Dim value As System.String

instance.Message = value

value = instance.Message
```

C#

```
System.string Message {get; set;}
```

C++/CLI

```
property System.String^ Message {  
    System.String^ get();  
    void set ( & System.String^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Non-empty message string

Visual Basic for Applications (VBA) Syntax

See [MessageBarDefinition::Message](#).

Remarks

This property must be set.

See Also

[IMessageBarDefinition Interface](#)
[IMessageBarDefinition Members](#)

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ResponseAText Property (IMessageBarDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMessageBarDefinition Interface](#) : ResponseAText Property (IMessageBarDefinition)

Gets or sets the text displayed on the first button or command link of this message bar.

.NET Syntax

Visual Basic (Declaration)

```
Property ResponseAText As System.String
```

Visual Basic (Usage)



```
Dim instance As IMessageBarDefinition  
Dim value As System.String
```

```
instance.ResponseAText = value
```

```
value = instance.ResponseAText
```

C#

```
System.string ResponseAText {get; set;}
```

C++/CLI

```
property System.String^ ResponseAText {  
    System.String^ get();  
    void set ( & System.String^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Text displayed on first user response control; default is an empty string

Visual Basic for Applications (VBA) Syntax

See [MessageBarDefinition::ResponseAText](#).

Example

See the [IMessageBarHandler](#) examples.

Remarks

This property must be set to a non-empty string if [IMessageBarDefinition::ResponseAType](#) is not [swUserMessageBarResponseType_e.swUserMessageBarResponseType_None](#).

See Also

[IMessageBarDefinition Interface](#)
[IMessageBarDefinition Members](#)

Availability

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ResponseAType Property (IMessageBarDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMessageBarDefinition Interface](#) : ResponseAType Property (IMessageBarDefinition)

Gets or sets the first response user control for this message bar.

.NET Syntax

Visual Basic (Declaration)

Property ResponseAType As System.Integer

Visual Basic (Usage)



```
Dim instance As IMessageBarDefinition  
Dim value As System.Integer
```

```
instance.ResponseAType = value
```

```
value = instance.ResponseAType
```

C#

```
System.int ResponseAType {get; set;}
```

C++/CLI

```
property System.int ResponseAType {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

First response user control as defined by [swUserMessageBarResponseType_e](#)

Visual Basic for Applications (VBA) Syntax

See [MessageBarDefinition::ResponseAType](#).

Example

See the [IMessageBarHandler](#) examples.

Remarks

Default value of this property is `swUserMessageBarResponseType_e.swUserMessageBarResponseType_None`.

See Also

[IMessageBarDefinition Interface](#)
[IMessageBarDefinition Members](#)

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ResponseBText Property (IMessageBarDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMessageBarDefinition Interface](#) : ResponseBText Property (IMessageBarDefinition)

Gets or sets the text displayed on the second button or command link of this message bar.

.NET Syntax

Visual Basic (Declaration)

Property ResponseBText As System.String

Visual Basic (Usage)



```
Dim instance As IMessageBarDefinition  
Dim value As System.String
```

```
instance.ResponseBText = value
```

```
value = instance.ResponseBText
```

C#

```
System.string ResponseBText {get; set;}
```

C++/CLI

```
property System.String^ ResponseBText {  
    System.String^ get();  
    void set ( & System.String^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Text displayed on second user response control; default is an empty string

Visual Basic for Applications (VBA) Syntax

See [MessageBarDefinition::ResponseBText](#).

Remarks

This property must be set to a non-empty string if [IMessageBarDefinition::ResponseBType](#) is not `swUserMessageBarResponseType_e.swUserMessageBarResponseType_None`.

See Also

[IMessageBarDefinition Interface](#)
[IMessageBarDefinition Members](#)

Availability

ResponseBType Property (IMessageBarDefinition)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IMessageBarDefinition Interface](#) : ResponseBType Property (IMessageBarDefinition)

Gets or sets the second response user control for this message bar.

.NET Syntax

Visual Basic (Declaration)

Property ResponseBType As System.Integer

Visual Basic (Usage)



```
Dim instance As IMessageBarDefinition
Dim value As System.Integer

instance.ResponseBType = value
value = instance.ResponseBType
```

C#

```
System.int ResponseBType {get; set;}
```

C++/CLI

```
property System.int ResponseBType {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Second response user control as defined by [swUserMessageBarResponseType_e](#)

Visual Basic for Applications (VBA) Syntax

See [MessageBarDefinition::ResponseBType](#).

Remarks

Default value of this property is swUserMessageBarResponseType_e.swUserMessageBarResponseType_None.

See Also

[IMessageBarDefinition Interface](#)
[IMessageBarDefinition Members](#)

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Severity Property (IMessageBarDefinition)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IMessageBarDefinition Interface](#) : Severity Property (IMessageBarDefinition)

Gets or sets the severity level of this message bar.

.NET Syntax

Visual Basic (Declaration)

Property Severity As System.Integer

Visual Basic (Usage)



```
Dim instance As IMessageBarDefinition
Dim value As System.Integer

instance.Severity = value
value = instance.Severity
```

C#

```
System.int Severity {get; set;}
```

C++/CLI

```
property System.int Severity {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Severity level as defined by [swUserMessageBarSeverity_e](#)

Visual Basic for Applications (VBA) Syntax

See [MessageBarDefinition::Severity](#).

Example

See the [IMessageBarHandler](#) examples.

Remarks

The default value of this property is swUserMessageBarSeverity_e.swUserMessageBarSeverity_Information.

See Also

[IMessageBarDefinition Interface](#)
[IMessageBarDefinition Members](#)

Availability

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Title Property (IMessageBarDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMessageBarDefinition Interface](#) : Title Property (IMessageBarDefinition)

Gets or sets the title of this message bar.

.NET Syntax

Visual Basic (Declaration)

```
Property Title As System.String
```

Visual Basic (Usage)



```
Dim instance As IMessageBarDefinition
Dim value As System.String
```

```
instance.Title = value  
value = instance.Title
```

C#

```
System.string Title {get; set;}
```

C++/CLI

```
property System.String^ Title {  
    System.String^ get();  
    void set ( &    System.String^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Non-empty title string

Visual Basic for Applications (VBA) Syntax

See [MessageBarDefinition::Title](#).

Example

See the [IMessageBarHandler](#) examples.

Remarks

This property must be set.

See Also

[IMessageBarDefinition Interface](#)
[IMessageBarDefinition Members](#)

Availability

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UniqueName Property (IMessageBarDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMessageBarDefinition Interface](#) : UniqueName Property (IMessageBarDefinition)

Gets the ID of this message bar.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property UniqueName As System.String
```

Visual Basic (Usage)



```
Dim instance As IMessageBarDefinition  
Dim value As System.String
```

```
value = instance.UniqueName
```

C#

```
System.string UniqueName {get;}
```

C++/CLI

```
property System.String^ UniqueName {  
    System.String^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Unique ID of this message bar

Visual Basic for Applications (VBA) Syntax

See [MessageBarDefinition::UniqueName](#).

Remarks

The ID is defined by the add-in in its call to [ISldWorks::DefineMessageBar](#).

It is the add-in's responsibility to ensure that the ID is unique by using, for example, a combination of add-in module name and unique message identifier:

"MyAddInName+ID_MYMESSAGE1"

See Also

[IMessageBarDefinition Interface](#)
[IMessageBarDefinition Members](#)

Availability

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